Scalability & Future Plan

Date	2 July, 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the OptiCrop system can be scaled to support larger user bases, increased agricultural datasets, and more advanced features. It also presents a roadmap for enhancing the system to support real-world agricultural decision-making at scale.

Current System Limitations:

S.No	**Limitation**	**Impact**	**Priority to Address(High / Medium / Low)**
1	No caching mechanism for repeated crop prediction requests	Increased response time for repeated queries	Medium
2	Single-server deployment (local system)	Limited ability to handle large numbers of concurrent users	High
3	Limited dataset coverage	Reduced prediction accuracy for diverse regions	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports around 100 concurrent users on a single instance	Deploy on cloud (AWS/Azure) with auto-scaling and load balancers to support 10,000+ concurrent users
2	Data Storage	Local dataset and storage	Migrate to scalable cloud storage (Amazon RDS / Azure SQL) with partitioning and backups
3	Model Efficiency	Single trained ML model	Implement model optimization and possibly ensemble or region-specific models
4	Security	Web-based interface only	Develop mobile application and API-based integration for wider access

Future Roadmap:

Phase	Planned Feature / Enhancement	Target timeline	Expected Impact
Phase-2	Integration with real-time weather APIs and soil sensors	Near Future	More accurate and dynamic crop recommendations
Phase-3	Multi-language support and mobile application	Near Future	Improved accessibility for farmers in different regions
Phase-4	AI-based yield prediction and fertilizer recommendation system	Future	Better farm planning and resource optimization